Endometriosis Early-Detection App

Project Plan: a symptom-tracking and care-linking platform for early recognition of endometriosis

Item	Detail
Project type	Mobile-friendly web app (Progressive Web App), later Android app
Primary users	Girls and women with painful or irregular periods; clinicians and facilities
Initial market	Kenya (English and Swahili)
Document date	10 October 2026
Status	Concept and planning stage

1. Vision, Problem and Goals

Problem. Endometriosis is often missed for years because severe period pain is normalised, symptoms are poorly documented, and many patients struggle to reach a clinician who suspects it. Late diagnosis leads to prolonged pain, fertility problems and lost school or work days.

Vision. A trusted, private app that helps users record their cycle and pain accurately, recognises patterns that warrant medical review, and connects them quickly to appropriate care.

Goals

- Help users log cycle, pain and related symptoms in under one minute a day.
- Flag patterns suggestive of endometriosis and prompt timely clinical review (screening support, not diagnosis).
- Generate a clinician-ready symptom report that speeds up consultations.
- Link users to gynaecologists, hospitals and clinics by location.
- Generate anonymised, consented data to support research on endometriosis in Africa.

Important principle. The app supports early recognition and referral. Only a qualified clinician can diagnose endometriosis, usually with examination, imaging and sometimes laparoscopy. All wording in the app must reflect this.

2. Target Users and Stakeholders

Group	Needs
Patients / users (adolescents and women)	Easy tracking, privacy, understandable results, access to care
Gynaecologists and doctors	Structured symptom history, referral requests, credibility
Clinics and hospitals	Listing, appointment requests, patient flow
Nurses and community health workers	Education tools and referral support
Universities and researchers	Ethical, anonymised data; validation studies
Funders and partners	Evidence of impact and sound governance

3. Feature Plan

Features are grouped by priority so the first version stays small and testable.

Priority	Feature	Description
MVP	Account and consent	Sign-up by phone or email, clear consent, optional PIN or biometric lock
MVP	Cycle tracker	Period start and end, flow level, cycle length, predictions, irregularity alerts
MVP	Pain log	0-10 score, body-map location, timing in cycle, effect on daily life

Priority	Feature	Description
MVP	Other symptoms	Pain with sex, urination or bowel movements, heavy bleeding, bloating, fatigue, nausea
MVP	Doctor report (PDF)	3-6 month summary of cycles and symptoms to take to the appointment
MVP	Education	Short, accurate articles on endometriosis, period pain and when to seek help
Phase 2	Pattern screening	Rule-based risk flag reviewed by gynaecologists; prompts to seek care
Phase 2	Facility directory	Search by county, specialty and services; contact and map details
Phase 2	Appointment requests	Send a request with the doctor report attached (with user consent)
Phase 2	Swahili and low-data mode	Full translation, offline logging, small download size
Phase 3	Telehealth	Video or chat consultations with partner clinicians
Phase 3	Peer community	Moderated support groups and anonymous Q&A with experts
Phase 3	SMS / USSD reminders	Reach users with basic phones
Phase 3	Research dashboard	Anonymised, consented data for approved studies

4. Symptom Screening Logic (Draft)

To be refined and approved by a gynaecologist. The aim is a simple, transparent set of rules, not a black-box score.

- Severe period pain (7 or more out of 10) in most cycles over 3 months.
- Pain that stops school, work or normal activity, or does not respond to usual painkillers.
- Pain during sex, urination or bowel movements around periods.
- Heavy bleeding or bleeding between periods.
- Difficulty conceiving, or a family history of endometriosis.
- Result wording: "Your pattern suggests you should speak to a doctor", never "you have endometriosis".
- Urgent warning: sudden severe pain, fainting or very heavy bleeding prompts immediate medical care.

5. Technical Plan

Area	Recommendation
Platform	Progressive Web App first (one codebase, works on any phone, installable); Android app later
Front end	React or Flutter, mobile-first, simple and calm design
Back end	Node.js or Python with a secure database (PostgreSQL), REST API
Security	Encryption in transit and at rest, role-based access, audit logs, regular backups
Hosting	Reputable cloud provider with strong security; consider data-location requirements

Area	Recommendation
Offline support	Local logging that syncs when the connection returns
Notifications	Push, SMS and later USSD reminders
Integrations	Maps for facilities; PDF generation; optional telehealth provider

6. Legal, Ethics and Privacy

- **Data protection:** comply with Kenya's Data Protection Act, 2019. Health data is sensitive personal data, so collect the minimum, obtain explicit consent, and register with the Office of the Data Protection Commissioner where required.
- **Clinical safety:** position the app as a screening and tracking aid, with medical disclaimers and emergency guidance.
- **Minors:** plan parental or guardian consent rules for users under 18.
- **Research ethics:** obtain approval from an Institutional Ethics Review Committee and, where required, NACOSTI before any study.
- **Medical-device rules:** check with the Pharmacy and Poisons Board whether any feature counts as regulated software.
- **Facility partners:** sign agreements covering data sharing and referral handling.
- **Content:** all medical content reviewed and signed off by a gynaecologist.

7. Roadmap and Timeline

Phase	Timeframe	Key activities	Outcome
0. Foundation	Months 1-2	Literature review; interviews with 10-15 women and 3-5 clinicians; recruit a gynaecologist adviser; define MVP	Validated problem and feature list
1. Design	Months 2-3	Wireframes, symptom questionnaire, branding, consent and privacy design	Clickable prototype
2. MVP build	Months 3-5	Build tracker, pain log, doctor report, education content	Working MVP
3. Pilot	Months 6-8	Test with 50-100 users, with ethics approval; collect feedback and usability data	Pilot report and improvements
4. Care linking	Months 8-10	Facility directory, appointment requests, partner agreements	Linked care pathway
5. Validation	Months 10-14	Compare screening flags with clinical outcomes; prepare publication	Evidence for funders and journals
6. Scale	Month 14+	Android app, telehealth, SMS/USSD, more counties	Wider rollout

8. Team and Partners

Role	Responsibility	How to fill
Project lead (you)	Vision, coordination, community and clinical liaison	Founder
Gynaecologist adviser	Clinical logic, content review, referral network	University or hospital partner
Developer	Build and maintain app	Student developer, freelancer or tech club
UI/UX designer	Simple, accessible design	Student or freelance
Data and ethics adviser	Privacy, research approvals, analysis	University supervisor
Community champions	User testing and awareness	Nursing students and clubs

Possible partners: your university and its research or innovation clubs, Kenyan hospitals and gynaecology departments, KEMRI, Amref, endometriosis and women's health NGOs, county health departments and ICT hubs.

9. Budget Estimate (Indicative)

Figures are rough planning estimates in Kenya shillings (KES) and should be replaced with real quotes. Many costs can fall sharply if students volunteer or a university supports the project.

Item	Estimate (KES)
Design and prototyping	30,000 - 80,000
Development of MVP	150,000 - 500,000
Hosting, domain, security (year 1)	40,000 - 120,000
Clinical review and content	30,000 - 100,000
Pilot study (data collection, incentives, ethics fees)	60,000 - 200,000
Legal and data-protection advice	30,000 - 100,000
Marketing and awareness	30,000 - 100,000
Approximate total, year 1	**370,000 - 1,200,000**

10. Funding Strategy

- University innovation or research grants and incubators.
- Women's health and digital health grants from foundations and development partners.
- Student and youth innovation competitions.
- Fellowships that fund research, such as the women's health fellowships discussed earlier.
- Partnerships with hospitals or NGOs that may co-fund pilots.
- Later: sustainable revenue through facility listings, premium reports or institutional licences, never by selling user data.

11. Success Measures

Area	Measure
Adoption	Number of registered users; users logging at least weekly
Engagement	Average number of logged cycles per user; report downloads
Clinical value	Users who sought care after a prompt; time from first symptoms to consultation
Accuracy	Agreement between screening flags and clinician assessment (validation study)
Satisfaction	User and clinician feedback scores
Research	Abstracts, publications or conference presentations

12. Risks and Mitigation

Risk	Mitigation
Misinterpreted as a diagnosis	Careful wording, disclaimers, clinician-reviewed rules
Data breach or misuse	Encryption, minimal data, strict access control, legal advice
Low user engagement	Quick logging, reminders, useful reports, community outreach
Limited internet or smartphones	Offline mode, low data use, SMS/USSD later
Lack of clinician buy-in	Involve gynaecologists from the start; co-design the report
Funding gaps	Phase the build; start with a low-cost MVP and student support
Stigma around menstrual health	Sensitive design, discreet app name and notifications

13. Immediate Next Steps (First 30 Days)

- Write a one-page concept note and share it with a lecturer or supervisor.
- Identify and approach a gynaecologist willing to advise.
- Interview at least 10 women about their period-pain experiences and care journeys.
- Review existing apps and research on endometriosis symptom tracking.
- Draft the symptom questionnaire and the doctor report layout.
- Sketch the main screens on paper, then build a clickable prototype.
- Look for a developer or tech club partner and a university innovation office.
- Identify the first grants or competitions to apply for.

This plan is a starting framework. Costs, timelines and regulatory requirements should be confirmed with the relevant professionals and authorities before launch.